SC Honors Geometry - Nappier

4.6 Similarity and Transformations

Speak **L**ike **A** **M**athematician:

similarity transformation:

similar figures:

Similarity Transformations

A _____ _____ is a dilation, or a composition of rigid motions and a dilation.

Two figures are _____ if and only if a similarity transformation maps one onto the other.

Notation: △ABC ~ △DEF is read "triangle ABC is _____ to triangle DEF."

What Similar Figures Share

Corresponding angles are _____.

Corresponding sides are _____ — their ratios all equal the _____ _____.

Congruent vs. Similar

Every congruence transformation is a similarity transformation with k = _____.

The reverse is NOT true: similar figures are congruent only when k = _____.

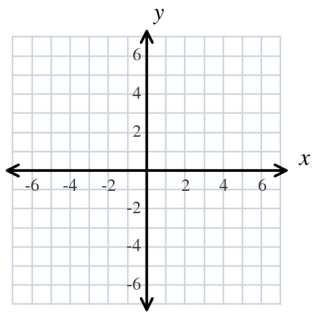
Example 1 – Name the Similarity Transformation

△ABC has A(2, 2), B(4, 6), and C(6, 2). Its image △A′B′C′ has A′(1, 1), B′(2, 3), and C′(3, 1).

a) Compare matching coordinates. What is the scale factor?

b) Name the similarity transformation.

c) Are the triangles congruent, similar, or both?



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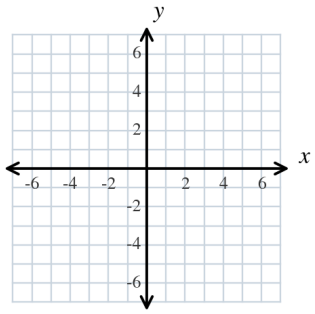
Example 2 – Dilation, Then Translation

P(1, 2) is dilated about the origin with k = 2, then translated along ⟨3, −1⟩.

a) Find the coordinates after the dilation.

b) Find the coordinates after the translation.

c) Is the composition a similarity transformation? Explain.



Example 3 – Reflection, Then Dilation

Q(2, −1) is reflected in the y-axis, then dilated about the origin with k = 3.

a) Find the coordinates after the reflection.

b) Find the coordinates after the dilation.

c) Does the order matter here? Try the dilation first.

Example 4 – Dilation, Then Rotation

R(2, 1) is dilated about the origin with k = 3, then rotated 90° about the origin.

a) Find the coordinates after the dilation.

b) Apply (a, b) ${to} (${M}b, a).

c) Which property is preserved, and which is not?